作业来自

梁昆淼《力学》 第 4 版 高等教育出版社

Johns Analytical Mechanics 2nd ed Oxford University Press

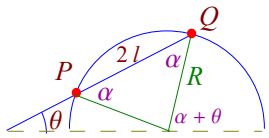
例如： L01.08 表示梁昆淼书第 01 章第 08 题； J02.05 Johns 书第 02 章第 05 题

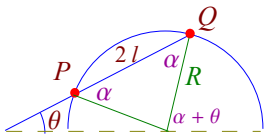
J03.07† 表示 Johns 书第 03 章第 07 题，带 † 表示仅要求荣誉班同学完成

Homework 05, Week 07 2023.10.16 Due 2023.10.23 L02.05, L02.09, J03 .07†, J03.08, J03.09, J03.10

★ L02.05

光滑钢丝弯成半径为 R 的半圆竖直放置，钢丝上穿着两个质点，
其重量分别为 P 和 Q，两质点由 长度为 $2l$ 的不可伸长
轻绳连起来，求平衡时绳与水平面的夹角，
如图（图参照教材 p442）





$$\alpha = \cos^{-1}(l/R)$$

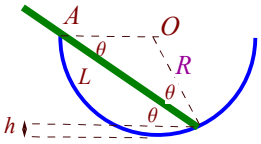
$$\begin{cases} y_P = R\sin(\alpha-\theta) \\ y_Q = R\sin(\alpha+\theta) \end{cases}$$

势能： $U = P\,R\sin(\alpha-\theta) + Q\,R\sin(\alpha+\theta)$

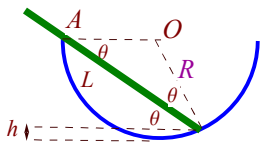
达朗贝尔原理： $\dfrac{dU}{d\theta} = 0 \implies -P\cos(\alpha-\theta) + Q\cos(\alpha+\theta) = 0 \implies \tan\theta = \dfrac{(Q-P)}{(Q+P)}\cot\alpha = \dfrac{(Q-P)}{(Q+P)}\dfrac{l}{\sqrt{R^2-l^2}}$

★ L02.09

光滑半球碗是固定的，匀质杆长 L 斜靠在碗边，
杆的下端为碗所承托，
上端则伸出碗外，试求杆的平衡位置。
在怎样条件下才有可能平衡？



设半球半径为 R，杆与水平方向的夹角为 θ，如图



$$U = M\,g\left(\frac{L}{2}\sin\theta + h\right), \quad h = R - R\sin 2\theta$$

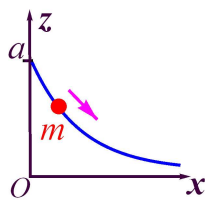
$$\implies U = M\,g\left(\frac{L}{2}\sin\theta + R - R\sin 2\theta\right)$$

达朗贝尔原理： $\dfrac{dU}{d\theta} = 0 \implies 8R\cos^2\theta - L\cos\theta - 4R = 0 \implies \cos\theta = \dfrac{L + \left(L^2 + 128R^2\right)^{1/2}}{16R}$

可能平衡的条件：$\cos\theta \le 1 \implies L \le 4R$

★ J03.08

A bead of mass m slides without friction on a rigid wire that lies in the xoz plane and has the shape $z = a\,e^{-\gamma x}$, where γ is some given constant. Gravity acts downwards, with $\vec{g} = -g\,\vec{e}_z = -g\,\vec{e}_3$.



(a) Write the full Lagrangian for this problem, and write the equations for the two holonomic constraints (几何约束), first that the mass is confined to the plane $y = 0$, and second that it is confined to the surface $z = a\,e^{-\gamma x}$.

(b) Write the three Lagange equations, intoducing the Lagrangian multipliers λ_1 and λ_2 as appropriate.

(c) Use the constraints to write a reduced Lagrangian $\tilde{L}(x, \dot{x}, t)$, with x serving as the single free coordinate.

Derive the reduced generalized energy from this reduced Lagrangian, and use it to find an expression for $\dot{x}^2$ as a function of x. (Assume that the mass is released from rest at the point $x = 0$.) Also use the reduced Lagrange equation to find an expression for $\ddot{x}$ as a function of x and $\dot{x}$

(d) Write an expression for the Cartesian vector force of the constraint $\vec{F}^{(\text{cons})}$ acting on the bead, expressing it as a function of x only. Check the limit of $\vec{F}^{(\text{cons})}$ as $x \to \infty$. Is it reasonable?

(a)

$$L = \frac{1}{2} m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - m\,g\,z, \quad G_1 = y, \quad G_2 = z - a\,e^{-\gamma x}$$

(b)

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = \lambda_1 \frac{\partial G_1}{\partial x} + \lambda_2 \frac{\partial G_2}{\partial x} \implies m\,\ddot{x} = \lambda_2\, a\,\gamma\, e^{-\gamma x}$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{y}} - \frac{\partial L}{\partial y} = \lambda_1 \frac{\partial G_1}{\partial y} + \lambda_2 \frac{\partial G_2}{\partial y} \implies m\,\ddot{y} = \lambda_1$$

with constraint $G_1 = y = 0$, one readily gets $y = 0$, $\dot{y} = 0$, $\ddot{y} = 0 \implies \lambda_1 = 0$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{z}} - \frac{\partial L}{\partial z} = \lambda_1 \frac{\partial G_1}{\partial z} + \lambda_2 \frac{\partial G_2}{\partial z} \implies m\,\ddot{z} + m\,g = \lambda_2$$

With the constraint $z = a\,e^{-\gamma x}$, one gets

$$\begin{cases} m\,\ddot{x} = \lambda_2\,\gamma\,z \\ m\,\ddot{z} + m\,g = \lambda_2 \end{cases} \qquad (1)$$

(c)

using constraints $\begin{cases} y = 0 \implies \dot{y} = 0 \\ z = a\,e^{-\gamma x} \implies \dot{z} = -a\,\gamma\,e^{-\gamma x}\,\dot{x} \end{cases}$ one gets the reduced Lagrangian

$$\tilde{L}(x, \dot{x}, t) = \frac{1}{2} m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - m\,g\,z = \frac{1}{2} m\,(\dot{x}^2 + a^2\,\gamma^2\,e^{-2\gamma x}\,\dot{x}^2) - m\,g\,a\,e^{-\gamma x}$$
$$= \frac{1}{2} m\,\dot{x}^2(1 + a^2\,\gamma^2\,e^{-2\gamma x}) - m\,g\,a\,e^{-\gamma x}$$

Lagrangian equation : $\dfrac{d}{dt}\dfrac{\partial \tilde{L}}{\partial \dot{x}} - \dfrac{\partial \tilde{L}}{\partial x} = 0$

$$\implies \frac{d}{dt}\left[m\,\dot{x}\,(1 + a^2\,\gamma^2\,e^{-2\gamma x})\right] + m\,\dot{x}^2\,a^2\,\gamma^3\,e^{-2\gamma x} - m\,g\,a\,\gamma\,e^{-\gamma x} = 0$$

$$\implies \ddot{x}\,(1 + a^2\,\gamma^2\,e^{-2\gamma x}) - 2\,\dot{x}^2\,a^2\,\gamma^3\,e^{-2\gamma x} + \dot{x}^2\,a^2\,\gamma^3\,e^{-2\gamma x} - g\,a\,\gamma\,e^{-\gamma x} = 0$$

$$\implies \ddot{x} = \frac{\dot{x}^2\,a^2\,\gamma^3\,e^{-2\gamma x} + g\,a\,\gamma\,e^{-\gamma x}}{(1 + a^2\,\gamma^2\,e^{-2\gamma x})} \qquad (2)$$

$$\tilde{H} = \frac{\partial \tilde{L}}{\partial \dot{x}}\dot{x} - \tilde{L} = \frac{1}{2} m \dot{x}^2 \left(1 + a^2 \gamma^2 e^{-2\gamma x}\right) + m g a e^{-\gamma x}$$

$$\frac{d\tilde{H}}{dt} = \frac{\partial \tilde{L}}{\partial t} = 0 \quad \Longrightarrow \quad \tilde{H} = \tilde{H}(t=0) = m g a$$

$$\Longrightarrow \quad \frac{1}{2} m \dot{x}^2 \left(1 + a^2 \gamma^2 e^{-2\gamma x}\right) + m g a e^{-\gamma x} = m g a$$

$$\Longrightarrow \quad \dot{x}^2 = \frac{2 g a (1 - e^{-\gamma x})}{\left(1 + a^2 \gamma^2 e^{-2\gamma x}\right)} \qquad (3)$$

(*d*)

$$F_x^{(\text{cons})} = \lambda_2 \frac{\partial G_2}{\partial x} = \lambda_2 a \gamma e^{-\gamma x}, \quad F_z^{(\text{cons})} = \lambda_2 \frac{\partial G_2}{\partial z} = \lambda_2$$

$\vec{F} = \lambda_2 \left(a \gamma e^{-\gamma x} \vec{e}_x + \vec{e}_z\right)$ where, from Equation (1), $\lambda_2 = \frac{m}{\gamma} \frac{\ddot{x}}{z} = \frac{m}{a \gamma} \ddot{x} e^{\gamma x}$

$\Longrightarrow$ With Equations (2) and (3) one gets the constraint force as a function of x

As $x \to \infty$, $\vec{F} = \lambda_2 \vec{e}_z = m g \vec{e}_z$

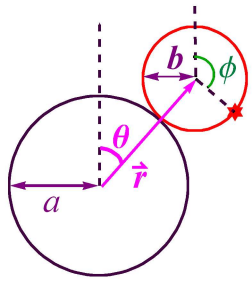
It is reasonable since the wire turns horizontal and the only support by the wire is against the gravity.

★ J03.09

A fixed, right circular cylinder (first cylinder) of radius a lies on its side, with its symmetry axis horizontal. A hollow right circular cylinder (second cylinder) of radius b and mass m is free to roll without slipping on the first one. Assume that its symmetry axis ramains aligned with that of the first cylinder. The full Lagragian for the second cylinder 's motion is

$$L = \frac{1}{2} m \left(\dot{r}^2 + r^2 \dot{\theta}^2\right) + \frac{1}{2} m b^2 \dot{\phi}^2 - m g r \cos\theta \qquad (1)$$

where r is the distance between the axes of the two cylinders, and θ and ϕ are angles shown in the figure.



Notice that ϕ is the angle between vertical and a mark ✱ inscribed on the face of the second cylinder.

(*a*) Write the two constraint functions, G_1 expressing the constraint that second cylinder is in contact with the first one, and G_2 expressing the condition of rolling without slipping. Assume that $\phi = 0$ when $\theta = 0$

(*b*) Write a (completely) reduced Lagragian $\tilde{L}\left(\theta, \dot{\theta}, t\right)$ and use the reduced generalized energy theorem to express $\dot{\theta}$ as a function of θ. Assume the second cylinder to be initially at rest, and at a very small distance to the right of $\theta = 0$.

(c) Use the Lagrangian given in Eq. (1) to write the three Lagrange equations, introducing Lagrange multipliers as appropriate. Find the generalized force of constraint $Q_r^{(\text{cons})}$ for the r variable and use it to find the angle θ_c at which the rolling cylinder will lose contact with the fixed one.

(*a*)

$$\begin{cases} G_1 = r - (a+b) \\ G_2 = (a+b)\,\theta - b\,\phi \end{cases}$$

注意 这里 $G_2 = (a+b)\,\theta - b\,\phi$

而非： $G_2 = a\,\theta - b\,\phi$

因为 B 点转到 C 处时

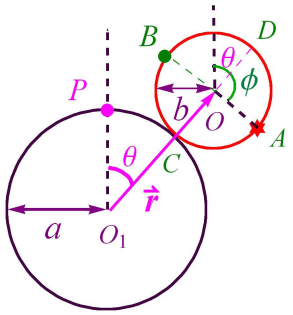
OA 点为竖直方向

$\overset{\frown}{\mathrm{PC}} = \overset{\frown}{\mathrm{BC}}$

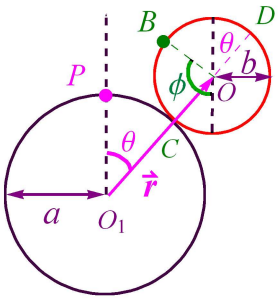
$\overset{\frown}{\mathrm{PC}} = a\,\theta,\quad \overset{\frown}{\mathrm{BC}} = b\,\angle\mathrm{COB}$

$\angle\mathrm{COB} = \angle\mathrm{AOD} = \phi - \theta$

故： $a\,\theta = b(\phi - \theta)$



或者简单一点：如下图，设开始时 B 点与 P 点接触， OB 与竖直方向的夹角为 0



经无滑动滚动到如图所示处，

OB 与竖直方向的夹角为 ϕ

无滑动滚动，故弧长 $\overset{\frown}{\mathrm{BC}} = \overset{\frown}{\mathrm{CP}}$

$\overset{\frown}{\mathrm{CP}} = a\,\theta,\quad \overset{\frown}{\mathrm{BC}} = b\,(\phi - \theta)$

故有： $a\,\theta = b\,(\phi - \theta)$

即： $(a+b)\,\theta - b\,\phi = 0$

关键在于圆柱 b 转了 ϕ 角度

但 $\angle\mathrm{COB} = \phi - \theta$

(*b*)

From $G_1 = 0 \implies r = a + b \implies \dot{r} = 0,\quad \ddot{r} = 0$

From $G_2 = 0 \implies \theta = \dfrac{b\,\phi}{a+b} \implies \dot{\theta} = \dfrac{b\,\dot{\phi}}{a+b}$ and $\dot{\phi} = \dfrac{(a+b)\,\dot{\theta}}{b}$

$$\tilde{L} = \frac{1}{2}\,m\,(a+b)^2\,\dot{\theta}^2 + \frac{1}{2}\,m\,b^2\,\dot{\phi}^2 - m\,g\,(a+b)\cos\theta$$

第一项为质心动能（轨道动能），第二项为内动能（自旋动能）

第三项为重力势能，这里利用了 Konig 定理

利用 $\dot{\phi} = \dfrac{(a+b)\,\dot{\theta}}{b}$ 消去受限坐标 ϕ 得

$$\tilde{L} = m\,(a+b)^2\,\dot{\theta}^2 - m\,g\,(a+b)\cos\theta$$

(*c*)

$$L = \frac{1}{2}\,m\left(\dot{r}^2 + r^2\,\dot{\theta}^2\right) + \frac{1}{2}\,m\,b^2\,\dot{\phi}^2 - m\,g\,r\cos\theta$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{r}} - \frac{\partial L}{\partial r} = \lambda_1\frac{\partial G_1}{\partial r} + \lambda_2\frac{\partial G_2}{\partial r} \implies m\,\ddot{r} - m\,r\,\dot{\theta}^2 + m\,g\cos\theta = \lambda_1$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = \lambda_1\frac{\partial G_1}{\partial \theta} + \lambda_2\frac{\partial G_2}{\partial \theta} \implies \frac{d}{dt}\left(m\,r^2\,\dot{\theta}\right) - m\,g\,r\sin\theta = \lambda_2(a+b)$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}} - \frac{\partial L}{\partial \phi} = \lambda_1\frac{\partial G_1}{\partial \phi} + \lambda_2\frac{\partial G_2}{\partial \phi} \implies m\,b^2\,\ddot{\phi} = -\lambda_2\,b$$

还有一个方程，来自广义能量定理，先求广义能量函数

$$\tilde{H} = \frac{\partial \tilde{L}}{\partial \dot{\theta}} \dot{\theta} - \tilde{L} = m\,(a+b)^2\,\dot{\theta}^2 + m\,g\,(a+b)\cos\theta = m\,g\,(a+b)$$

where one has used $\dfrac{d\tilde{H}}{dt} = -\dfrac{\partial \tilde{L}}{\partial t} = 0 \quad \Longrightarrow \quad \tilde{H} = \tilde{H}(t=0) = m\,g\,(a+b)$

$$\Longrightarrow \quad \dot{\theta}^2 = \frac{g\,(1-\cos\theta)}{(a+b)}$$

The generalized force of constraint

$$Q_r^{(\text{cons})} = \lambda_1 \frac{\partial G_1}{\partial r} + \lambda_2 \frac{\partial G_2}{\partial r} = \lambda_1 = m\,\ddot{r} - m\,r\,\dot{\theta}^2 + m\,g\cos\theta$$

but $\ddot{r} = 0$ due to constraint, so $Q_r^{(\text{cons})} = -m\,(a+b)\,\dot{\theta}^2 + m\,g\cos\theta = m\,g\cos\theta - m\,g(1-\cos\theta)$

$$Q_r^{(\text{cons})} = m\,g\,(2\cos\theta - 1)$$

$$Q_r^{(\text{cons})} \geq 0 \quad \Longrightarrow \quad \cos\theta \geq \frac{1}{2} \quad \Longrightarrow \quad \theta \leq \frac{\pi}{3} = \theta_c$$

★ J03.10

Consider a bead of mass m sliding without friction on a wire in the x-o-z plane bent into shape of an inverted parabola $z = a - b\,x^2$. At time zero, the bead is at $x = 0$, $z = a$ with a velocity $\vec{v}_0 = v_0\,\vec{e}_x = v_0\,\vec{e}_1$. Constants a, b, v_0 are all greater that zero. Gravity acts in the negative z direction.

(a) Write an expression for the vector constraint force as a function of x only.

(b) Find the value of v_0 as a function of g, a, b which will make the constraint force exactly zero for all time $t > 0$.

约束函数：$G_1 = z - a + b\,x^2$

拉格朗日量：$L = \dfrac{1}{2} m\left(\dot{x}^2 + \dot{z}^2\right) - m\,g\,z$

完整的拉格朗日方程：$\dfrac{d}{dt}\dfrac{\partial L}{\partial \dot{x}} - \dfrac{\partial L}{\partial x} = \lambda_1 \dfrac{\partial G_1}{\partial x}$

$$\begin{cases} m\,\ddot{x} = 2\,\lambda_1\,b\,x \\ m\,\ddot{z} + m\,g = \lambda_1 \end{cases}$$

约化拉格朗日量：消去受限坐标 z：$z = a - b\,x^2$，$\dot{z} = -2\,b\,x\,\dot{x}$

$$\tilde{L} = \frac{1}{2} m\left(1 + 4\,b^2\,x^2\right)\dot{x}^2 - m\,g\,a + m\,g\,b\,x^2$$

约化的拉格朗日方程：$\dfrac{d}{dt}\dfrac{\partial \tilde{L}}{\partial \dot{x}} - \dfrac{\partial \tilde{L}}{\partial x} = 0$

化简得：$\left(1 + 4\,b^2\,x^2\right)\ddot{x} = 2\,g\,b\,x - 4\,b^2\,x\,\dot{x}^2$

约化的广义能量函数：

$$\tilde{H} = \dot{x}\,\frac{\partial \tilde{L}}{\partial \dot{x}} - \tilde{L} = \frac{1}{2} m\left(1 + 4\,b^2\,x^2\right)\dot{x}^2 + m\,g\,a - m\,g\,b\,x^2$$

广义能量定理：$\dfrac{d\tilde{H}}{dt} = -\dfrac{\partial \tilde{L}}{\partial t} = 0$，　$\tilde{H}$ 守恒

代入 $\tilde{H}$ 的初值：$\tilde{H}_0 = \dfrac{1}{2} m\,v_0^2 + m\,g\,a$

故：$\tilde{H} = \tilde{H}_0 \implies \dot{x}^2 = \dfrac{v_0^2 + 2\,g\,b\,x^2}{1 + 4\,b^2\,x^2}$ 代入绿色式

$$\ddot{x} = \frac{2\,b\,x}{(1 + 4\,b^2\,x^2)^2}\,(g - 2\,b\,v_0^2)$$

$$\implies \lambda_1 = \frac{m\,\ddot{x}}{2\,b\,x} = \frac{m\,(g - 2\,b\,v_0^2)}{(1 + 4\,b^2\,x^2)^2}$$

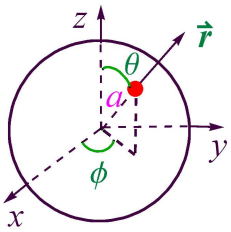
$$\vec{F}^{(\text{cons})} = \lambda_1 \frac{\partial G_1}{\partial \vec{r}} = \lambda_1 \nabla G_1 = \frac{m\,(g - 2\,b\,v_0^2)}{(1 + 4\,b^2\,x^2)^2}\,(2\,b\,x\,\vec{e}_x + \vec{e}_z)$$

要使得 $\vec{F}^{(\text{cons})} = 0$，需要 $(g - 2\,b\,v_0^2) = 0$

这时 $z = a - b\,x^2$ 恰好与平抛轨迹相符合，当然无需约束力。

★ J03.07† （荣誉课程同学完成）

A single mass m slides without friction on the outside surface of a sphere of radius a. Suppose that at $t = 0$, it has spherical polar coordinates $\theta_0 > 0$, $\phi_0 = 0$ and generalized velocities $\dot{\theta}_0 = 0$ and $\dot{\phi}_0 > 0$.



(a) Using spherical polar coordinates, write both the full Lagrangian and the reduced Lagrangian for this porblem.

"full" 和 "reduced" 分别指拉格朗日中没有消去和消去受限坐标 $q^{(b)}$, $\dot{q}^{(b)}$ 情况。

(b) Write the Lagrange equations for both the full and the reduced Lagrangian

(c) Use the reduced Lagrangian $\tilde{L}$ to write the reduced generalized energy $\tilde{H}_q$

(d) The mass will leave the surface of the sphere at the instant at which the normal force of the constraint becomes negative. Write an expression for the angle $\theta_{\max}$ at which the mass leaves the sphere.

(e) Define the parameter γ by $a\,\dot{\phi}_0^2 = \gamma\,g$, where g is the acceleration of gravity. Express $\theta_{\max}$ as a function of a, g, γ, θ_0 only. In the case with $\gamma = 0$, find $\theta_{\max}$ in the limit of $\theta_0 \to 0$.

(f) With $\theta_0 = 10\,°$, find the numerical value of $\theta_{\max}$ for the case in which $\gamma = 0$. How big would γ have to be in order to reduce $\theta_{\max}$ to $45\,°$.

取球坐标，约束函数：$G_1 = r - a$

完整的拉格朗日量：

$$L = \frac{m}{2}\left(\dot{r}^2 + r^2\,\dot{\theta}^2 + r^2 \sin^2\theta\,\dot{\phi}^2\right) - m\,g\,r\cos\theta$$

约化的拉格朗日量：$\tilde{L} = \dfrac{m\,a^2}{2}\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\phi}^2\right) - m\,g\,a\cos\theta$

完整的拉格朗日方程：$\dfrac{d}{dt}\dfrac{\partial L}{\partial \dot{q}_k} - \dfrac{\partial L}{\partial q_k} = \lambda_1 \dfrac{\partial G_1}{\partial q_k}$

$q_1 = r$: $\quad m\,\ddot{r} - m\,r\,\dot{\theta}^2 - m\,r\sin^2\theta\,\dot{\phi}^2 + m\,g\cos\theta = \lambda_1$

$q_2 = \theta$: $\quad \dfrac{d\left(m\,r^2\,\dot{\theta}\right)}{dt} - \left[m\,r^2\sin\theta\cos\theta\,\dot{\phi}^2 - m\,g\,r\sin\theta\right] = 0$

注意这时第一项先不急着展开

$q_3=\phi$： $\dfrac{d\left(m r^2 \sin^2\theta\,\dot{\phi}\right)}{dt}=0$， ϕ 是可遗坐标

好不容易得到一个运动常数，不要展开

得：$m r^2 \sin^2\theta\,\dot{\phi}=m\ r_0^2 \sin^2\theta_0\,\dot{\phi}_0$

现在，可以把约束条件 $r=a$ 代入个方程进行简化了

$$\begin{cases} -m a\dot{\theta}^2-m a\sin^2\theta\,\dot{\phi}^2+m g\cos\theta=\lambda_1 \\ m a^2\ddot{\theta}-m a^2\sin\theta\cos\theta\,\dot{\phi}^2-m g a\sin\theta=0 \\ \sin^2\theta\,\dot{\phi}=\sin^2\theta_0\,\dot{\phi}_0 \end{cases}$$

从约化拉格朗日量也可得拉格朗日方程：

$$\begin{cases} m a^2\ddot{\theta}-m a^2\sin\theta\cos\theta\,\dot{\phi}^2-m g a\sin\theta=0 \\ \sin^2\theta\,\dot{\phi}=\sin^2\theta_0\,\dot{\phi}_0 \end{cases}$$

广义能量函数： $\tilde{H}=\dot{\theta}\dfrac{\partial\tilde{L}}{\partial\dot{\theta}}+\dot{\phi}\dfrac{\partial\tilde{L}}{\partial\dot{\phi}}-\tilde{L}$

$$\tilde{H}=\frac{1}{2}m a^2\left(\dot{\theta}^2+\sin^2\theta\,\dot{\phi}^2\right)+m g a\cos\theta$$

因为没有非保守力，$\dot{\tilde{H}}=-\dfrac{\partial\tilde{L}}{\partial t}=0$， $\tilde{H}$ 是运动常数，我喜欢

$$\frac{1}{2}m a^2\left(\dot{\theta}^2+\sin^2\theta\,\dot{\phi}^2\right)+m g a\cos\theta=\frac{1}{2}m a^2\left(\dot{\theta}_0^2+\sin^2\theta_0\,\dot{\phi}_0^2\right)+m g a\cos\theta_0$$

$$\dot{\theta}^2+\sin^2\theta\,\dot{\phi}^2+\frac{2g}{a}\cos\theta=\dot{\theta}_0^2+\sin^2\theta_0\,\dot{\phi}_0^2+\frac{2g}{a}\cos\theta_0$$

$$\dot{\theta}^2=\frac{2g}{a}(\cos\theta_0-\cos\theta)+\sin^2\theta_0\,\dot{\phi}_0^2-\sin^2\theta\,\dot{\phi}^2$$

$\vec{F}^{(\text{cons})}=\lambda_1\dfrac{\partial G_1}{\partial\vec{r}}=\lambda_1\nabla G_1=\lambda_1\dfrac{\vec{r}}{a}$， 即： $\boxed{\vec{F}^{(\text{cons})}=\lambda_1\dfrac{\vec{r}}{a}}$

而 λ_1 由完整的拉格朗日方程得（约化的拉格朗日方程得不到 λ_1）：

$$-m a\dot{\theta}^2-m a\sin^2\theta\,\dot{\phi}^2+m g\cos\theta=\lambda_1$$

将紫色方程代入：$\lambda_1=-2m g\cos\theta_0+3m g\cos\theta-m a\sin^2\theta_0\,\dot{\phi}_0^2$

题中定义了：$\gamma=a\dot{\phi}_0^2/g$ 故：$\lambda_1=m g(3\cos\theta-2\cos\theta_0)-m g\gamma\sin^2\theta_0$

$\theta_{\max}$ 使得 $\vec{F}^{(\text{cons})}=0$， 故 $\lambda_1\big|_{\theta=\theta_{\max}}=0\implies 3\cos\theta_{\max}=2\cos\theta_0-\gamma\sin^2\theta_0$

$$\theta_{\max}=\cos^{-1}\left[\frac{2\cos\theta_0+\gamma\sin^2\theta_0}{3}\right]$$

当 $\gamma=0$ 时，$\lim\limits_{\theta_0\to 0}\theta_{\max}=\cos^{-1}\dfrac{2}{3}$

当 $\theta_0=10^\circ$，$\gamma=0$ 时，$\theta_{\max}=\cos^{-1}\left[\dfrac{2\cos 10^\circ}{3}\right]=49^\circ$

为了使 $\theta_{\max}=45^\circ$，蓝色方程可解得 $\gamma=\dfrac{3\cos\theta_{\max}-2\cos\theta_0}{\sin^2\theta_0}$

代入 $\theta_0=10^\circ$ 和 $\theta_{\max}=45^\circ$ 求得：$\gamma=5.0^\circ$